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Decline in Incidence of Epidemic Glucose Intolerance in Nauruans: Implications for the "Thrifty Genotype"

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Trends in the prevalence and incidence of non-insulin-dependent diabetes mellitus and impaired glucose tolerance were studied in Micronesian Nauruans aged 20 years and above, by linking glucose tolerance data collected during population surveys performed in 1975/1976, 1982, and 1987. Based on World Health Organization criteria, the age-standardized prevalence of non-insulin-dependent diabetes mellitus has remained relatively constant (24.0% in 1987), but the prevalence of impaired glucose tolerance has decreased significantly from 21.1% (95% confidence interval (CI) 17.0–25.3) in 1975/1976 to 8.7% (95% CI 7.1–10.3) in 1987. Between the periods 1975/1976–1982 and 1982–1987, the incidence of progression from normal glucose tolerance to either impaired glucose tolerance (incidence rate ratio (IRR) = 0.55, $p < 0.01$) or non-insulin-dependent diabetes (IRR = 0.46, $p < 0.05$) has decreased dramatically, while progression from impaired glucose tolerance to non-insulin-dependent diabetes has increased (IRR = 1.23). The overall age-standardized incidence of non-insulin-dependent diabetes has declined from 26.2 cases/1,000 person-years in 1975/1976–1982 to 22.5 cases/1,000 person-years in 1982–1987. As there were no changes in the frequency of recognized risk factors, the decline in incidence of glucose intolerance is probably due to the intensity of the epidemic in Nauru, which has already removed a high proportion of the genetically susceptible individuals from the pool with normal glucose tolerance. Coupled with the observations that mortality is higher and fertility lower in diabetic Nauruans across the age range, the observed decline in the incidence and prevalence of non-insulin-dependent diabetes mellitus and impaired glucose tolerance may presage a fall in the population frequency of the diabetic genotype, at least in its more severe form, as might be predicted on the basis of Neel's "thrifty genotype" hypothesis. *Am J Epidemiol* 1991;133:1093–1104.

diabetes mellitus, non-insulin-dependent; genotype; glucose tolerance test; incidence; prevalence; selection (genetics)

The past 3–4 decades have witnessed virtual epidemics of non-insulin-dependent diabetes mellitus in many traditional populations as their life-styles have changed from hunting, gathering, and subsistence agriculture to a modernized pattern characterized by sedentary occupations and energy-dense processed foods. Numerous epidemiologic studies have highlighted the susceptibility of Micronesian and Polynesian Pacific islanders (1–3), North American Indians (4, 5), Asian Indians (6, 7), and Australian Aborig-

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Abbreviations. CI, confidence interval; IRR, incidence rate ratio.

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ines (8) to non-insulin-dependent diabetes mellitus. There is sufficient anecdotal and documentary evidence available to suggest that non-insulin-dependent diabetes mellitus either did not occur or, at the most, must have been exceedingly rare in these populations at the time of first contact with Europeans (and indeed for many years following contact) (9, 10).

The Micronesian population of the small central Pacific island of Nauru has the second highest recorded incidence and prevalence of non-insulin-dependent diabetes mellitus (2, 11). In 1987, the prevalence in adults aged 25–74 years, age-standardized to the Australian population, was 37 percent, 13 times that of European-origin Australians (12). The marked susceptibility to non-insulin-dependent diabetes mellitus of traditional populations, such as Nauruans, has been explained in terms of the "thrifty genotype" (13, 14). This hypothesis suggests that these populations were subject to periods of famine associated with events such as droughts, hurricanes, and long migratory canoe voyages, which gave a selective advantage to individuals whose metabolism stored energy with maximum efficiency. In modern times, when there is an assured supply of energy from refined foods and a sedentary life-style, this genotype has become disadvantageous and results in obesity, hyperinsulinemia, and insulin resistance, which lead ultimately to pancreatic beta cell decompensation and non-insulin-dependent diabetes mellitus (3).

We report here the results of a recent follow-up study of glucose tolerance in Nauruans and describe changes in the prevalence and incidence of non-insulin-dependent diabetes mellitus and impaired glucose tolerance relative to earlier reports (2, 9, 11, 15). These changes have implications for other populations that either already or may in the future experience similar "epidemic" glucose intolerance.

MATERIALS AND METHODS

Background

Nauru is a small (approximately 20-km circumference) isolated island in the central Pacific Ocean, just south of the equator. The indigenous population of approximately 5,500 persons is of Micronesian ancestry. The Nauruans have become extremely wealthy from the mining of rich phosphate deposits on the island, and their life-style is marked by a high consumption of imported processed foodstuffs, a predominantly sedentary activity pattern, and obesity. A pilot survey performed in May 1975 on 221 subjects aged 15 years and over who were resident in two of 12 districts of the island found the prevalence of diabetes to be 34.4 percent (70 percent of cases undiagnosed), using a plasma glucose of 160 mg/100 ml (8.9 mmol/liter) 2 hours following a 75-g oral glucose tolerance test, as the diagnostic criterion (15). A further screening survey conducted in January 1976 found a similar prevalence in 417 attenders aged 10 years and over (9). In January 1982, a more ambitious study was performed that had as its target population all Nauruans aged 20 years and over, with 1,561 attenders and an estimated response rate of 83 percent (2). This survey included 266 individuals who were not diabetic in the combined 1975/1976 surveys and allowed the incidence and natural history of glucose intolerance to be studied for the period 1975/1976–1982 (11).

Study sample

In February 1987, the entire Nauruan population was screened for hepatitis B antigen. We took this opportunity to also screen all individuals aged 15 years and over for diabetes. Subjects who had not attended one of the earlier surveys had a random or fasting plasma glucose determined. Those who had attended in 1975/1976 *and/or* 1982 underwent more complete investigation, including an oral glucose tolerance test in subjects not previously found to have diabetes. The latter group (subjects with normal or impaired glucose tolerance) form the basis of the present study.

The eligible population was all ethnic Nauruans aged 20 years and above, then alive and resident on the island (including those temporarily overseas), who had previously attended either or both the 1975/1976

and 1982 surveys. Only subjects with at least one parent identified as being of Nauruan heritage are included in these analyses. Because earlier reports included some non-Nauruans, prevalence and incidence for the comparisons in this paper have been recalculated, resulting in small differences in numbers from those originally published (2, 9, 11, 15). There were 1,497 eligible individuals, comprising 62 percent of the total resident Nauruan population aged ≥20 years and 78 percent of those aged ≥25 years, as enumerated at a census performed immediately prior to the survey.

The 1987 response rate in eligible subjects who were seen in both 1975/1976 and 1982 was 89 percent, similar to that of subjects seen in 1982 alone (88 percent), but greater than that in those seen in 1975/1976 but not in 1982 (69 percent). The overall response rate (subjects seen in either or both earlier surveys) was 86 percent. Of these individuals, 93 percent (n = 1,201) had classifiable glucose tolerance data after subjects who had incomplete glucose tolerance tests were excluded. The breakdown of the 1987 survey population is depicted in figure 1. The final group of respondents were considered to be representative of both the total and eligible populations in terms of known health indicators (16) and age (table 1), except that the 20- to 24-year-old age group was relatively underrepresented because of different age eligibility criteria at earlier surveys.

Nauruans form a small but proud and homogeneous population, and migration is very rare. The Nauru government estimates that less than 1 percent of the approximately 5,500 ethnic Nauruans live overseas permanently. Of the 1,467 subjects eligible for the 1987 follow-up, there were 207 nonresponders, of whom 67 were overseas at the time of the survey (i.e., 4 percent of eligible subjects). All of these subjects were either on holiday or regarded as living away from Nauru *temporarily*. Their characteristics at earlier surveys (mean age, mean body mass index, prevalence of impaired glucose tolerance, and prevalence of non-insulin-dependent diabetes mellitus, in particular) were practically identical to those of the responders and the larger nonresponder group. There is therefore no reason to believe that differential migration in either the total or eligible populations should influence the trends in incidence and prevalence of glucose intolerance that are examined in this study.

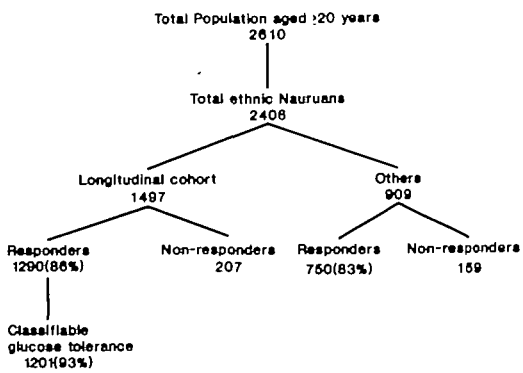

FIGURE 1. Flow chart indicating eligibility and response for 1987 follow-up survey in Nauru.

TABLE 1. Comparability of total, eligible, and responder populations for ethnic Nauruans: 1987 survey

Age group (years)	Total population		Eligible sample		Responders	
	n	%*	n	%	n	%
20–24	577		70		63	
25–34	939	51.3	667	46.7	589	48.0
35–44	396	21.7	334	23.4	280	22.8
45–54	229	12.5	200	14.0	174	14.2
55–64	182	10.0	158	11.1	134	10.9
65+	83	4.5	68	4.8	50	4.1
Total 25+	1,829		1,427		1,227	
20+	2,406		1,497		1,290	

* Percentage of population grouping aged 25+ years (i.e., excluding subjects aged 20–24 years). Subjects aged 20–24 years excluded from comparison because different age eligibility for 1975/1976 and 1982 surveys resulted in relatively few of this age group in the total population being eligible for the 1987 follow-up study.

Survey procedure

In 1987, all eligible subjects were asked to fast overnight and attend a central survey site the following day between 8 and 10 a.m. (16). Subjects who did not have a validated history of diabetes underwent a 2-hour oral glucose tolerance test using 75 g of glucose monohydrate dissolved in 200 ml of water. Fasting and 2-hour venous blood specimens were collected into heparinized fluoridated tubes and centrifuged immediately. Plasma glucose levels were determined on-site within 3 hours using a YSI Model 23AM glucose analyzer (Yellow Springs Instrument Co., Yellow Springs, Ohio) that utilizes a glucose oxidase method. This methodology was similar to that followed in 1982 (2) and 1975/1976 (9, 15) except that, in the latter case, plasma was stored frozen and glucose assays were performed subsequently in Melbourne, Australia, using a SMAC 20 autoanalyzer (Technicon Instruments Corporation, Tarrytown, New York). In 1987, every tenth 2-hour specimen was assayed in duplicate in the field (coefficient of variation = 1.2 percent) and subsequently on stored plasma by autoanalyzer (r = 0.99; mean difference, −0.13 mmol/liter; $p < 0.01$) in Melbourne. Similar results were obtained in 1982.

Classification of glucose tolerance status

Identical criteria, based on those suggested by the World Health Organization in 1985, have been used to classify glucose tolerance status at each survey point (17). Diabetes mellitus was diagnosed if the 2-hour plasma glucose concentration was ≥11.1 mmol/liter or, in the absence of a 2-hour specimen, if the fasting level was ≥7.8 mmol/liter. Diagnoses established at earlier surveys (World Health Organization, 1985 only) were accepted, based on the premise that the disease is lifelong. Impaired glucose tolerance was defined by a fasting plasma glucose concentration <7.8 mmol/liter, and 2-hour plasma glucose level ≥7.8 mmol/liter but <11.1 mmol/liter. Subjects with both fasting and 2-hour values <7.8 mmol/liter were classified as having normal glucose tolerance.

Other survey data

Subjects were interviewed at the time of each survey, when a standard questionnaire that was completed by trained interviewers included information on cigarette smoking habits and obstetric history. Current cigarette smoking (percent) and mean number of livebirths (by glucose tolerance) are reported here. Total physical activity (available only for 1975/1976 and 1987) was graded by interviewers using a four-level scale corresponding to sedentary, light, moderate, and heavy. For this study, the two lower categories have been combined and are described as "low" physical activity.

Weight and height were measured at each survey on subjects without shoes and in light clothing, using standard equipment. Body mass index was calculated as weight (kg)/height $(m)^2$. Frozen fasting plasma specimens were transported by air to Melbourne, where routine biochemistry was performed, including measurement of triglyceride levels.

Statistical analysis

Data from each of the three surveys were linked to trace the natural history of glucose intolerance in Nauruans. Prevalence of impaired glucose tolerance and non-insulin-dependent diabetes mellitus at each survey and incidence rates for the periods 1975/1976–1982 and 1982–1987 have been compared. Only subjects aged 20 years or over at the time of any survey have been included.

All estimates have been age standardized using the direct method (18) to the age distribution of the 1982 survey population. Age-specific and age-standardized incidence rates (expressed as "per 1,000 person-years") have been calculated by reference to the baseline age groups and by dividing the cumulative incidence by the relevant person-time period. Differences between age-adjusted incidence (density) rates, as well as summary incidence rate ratios and their test-based 95 percent confidence intervals (CI), have been assessed using the Mantel-Haenszel techniques described by Kleinbaum et al. (19).

There were 830 nondiabetic subjects in 1982 who had valid follow-up glucose tolerance information in 1987. This group included 143 subjects who were also nondiabetic in 1975/1976. As age-specific and total incidence rates for 1982–1987 for the latter group alone were very similar to those of the larger cohort of 687 subjects who had baseline data only for 1982, results in this paper are given only for the total group of 830 subjects.

RESULTS

Prevalence

The prevalence of non-insulin-dependent diabetes mellitus increased dramatically with age at each of the surveys, although the shape of the curve varied between surveys in older age groups (figure 2). The age-specific prevalence of non-insulin-dependent diabetes was similar to or higher in 1975/1976 than in either 1982 or 1987, up until the age of 45–54 years, after which the 1975/1976 prevalence declined. The 1982 and 1987 prevalences were similar at all ages except that, in subjects aged 65 years and above, prevalence leveled off in 1987 but declined in 1982. Reflecting these overall patterns, the age-standardized prevalence of non-insulin-dependent diabetes was higher in 1975/6 (27.9 percent) versus 1982 (24.7 percent) and 1987 (24.0 percent). At any survey, the prevalences in males and females were similar (table 2).

By contrast to the pattern seen for non-insulin-dependent diabetes, the age-specific and age-standardized prevalence of impaired glucose tolerance was markedly lower in 1987 (8.7 percent) than in either 1975/1976 (21.1 percent) or 1982 (17.4 percent) (table 2; figure 3). Again, there was little difference between the sexes.

Incidence

The age-standardized incidence of non-insulin-dependent diabetes for subjects with either normal or impaired glucose tolerance at baseline was 14 percent lower in the second period: 26.2 cases/1,000 person-years for 1975/1976–1982 versus 22.5 cases/1,000 person-years for 1982–1987. This difference was not statistically significant ($p = 0.22$). On an age-specific basis, the 1975/1976–

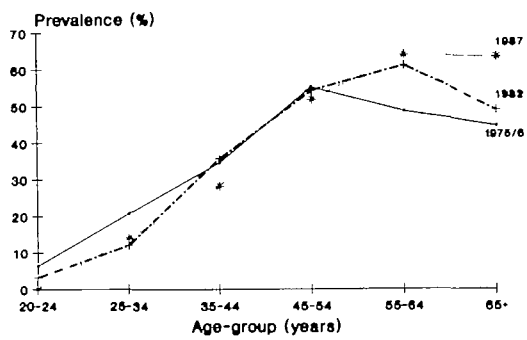

FIGURE 2. The age-specific prevalence of non-insulin-dependent diabetes mellitus at 1975/1976, 1982, and 1987 surveys in Nauru (sexes combined).

TABLE 2. Age-standardized* prevalence of NIDDM† and IGT† at 1975/1976, 1982, and 1987 surveys in Nauru

	1975/1976			1982			1987		
	n	Prevalence	95% CI†	*n*	Prevalence	95% CI	*n*	Prevalence	95% CI
Males	203			662			531		
IGT		17.2	11.6–22.7		17.5	14.6–20.4		7.6	5.6–9.7
NIDDM		30.1	24.0–36.1		24.7	21.9–27.5		23.8	20.9–26.7
Females	218			744			670		
IGT		24.9	18.9–30.9		17.2	14.5–19.8		9.5	7.2–11.8
NIDDM		25.9	20.4–31.3		24.6	21.8–27.3		24.1	21.5–26.6
Combined	421			1,406			1,201		
IGT		21.1	17.0–25.3		17.4	15.4–19.3		8.7	7.1–10.3
NIDDM		27.9	23.8–32.0		24.7	22.7–26.6		24.0	22.1–25.9

* Standardized using the direct method to the 1982 survey population.
† NIDDM, non-insulin-dependent diabetes mellitus; IGT, impaired glucose tolerance; CI, confidence interval.

1982 incidence increased sharply until the 45- to 54-year group and then declined in the ≥55-year group (only 10 subjects) (figure 4). Conversely, in 1982–1987, there were a stabilization and then continued increase in incidence in subjects aged 55 years and above. Confidence intervals, however, overlapped at these upper age intervals.

Natural history

The similarity of the overall incidence rates for non-insulin-dependent diabetes between the two survey periods conceals some interesting changes in the natural history of glucose intolerance. There was a 48 percent or greater fall in the age-standardized rate of progression from normal to either impaired glucose tolerance (41.4-21.6 cases/1,000 person-years; $p < 0.01$) or non-insulin-dependent diabetes (17.1-7.4 cases/1,000 person-years; $p < 0.05$) between 1975/1976–1982 and 1982–1987, respectively (table 3). Conversely, the rate of progression from impaired glucose tolerance to non-insulin-dependent diabetes increased from 35.2 to 56.1 cases/1,000 person-years ($p = 0.44$), and the rate of reversion from impaired glucose tolerance to normal increased from 69.8 to 88.1 cases/1,000 person-years ($p = 0.32$).

These changes in the natural history of glucose intolerance resulted in an overall decline in the incidence and prevalence of impaired glucose tolerance, a stabilization in the prevalence of non-insulin-dependent diabetes, and a decline in the overall incidence of non-insulin-dependent diabetes.

Risk factor levels

To investigate the possibility that improvements in life-style might be responsible for the incidence changes, risk factor levels have been determined for nondiabetic Nauruans at each survey (table 4). There is no indication that there have been beneficial changes in the levels of obesity (as reflected by the body mass index) or physical activity, the two most widely recognized modifiable risk factors for non-insulin-dependent diabetes. Similarly, there is little evidence for improvements in diet or general health behavior, based on mean triglyceride and total cholesterol levels, which increased between 1982 and 1987. Although cigarette smoking did decline significantly between 1975/1976 and 1987 in subjects with normal glucose tolerance, the pattern was reversed in the impaired glucose tolerance group, and for the two groups pooled, there was little difference between the surveys.

Glucose tolerance and fertility

Data on mean number of livebirths according to age group and glucose tolerance status, as determined at the 1987 survey, are presented in table 5. Diabetic women of each age group have a lower number of livebirths than do women with normal glucose toler-

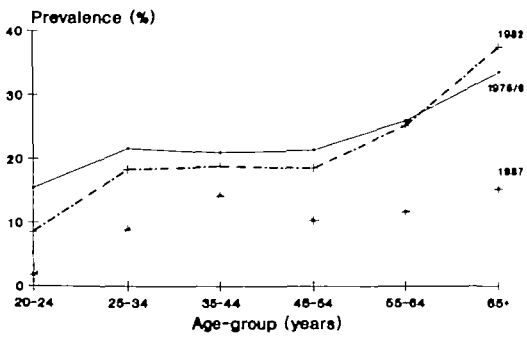

FIGURE 3. The age-specific prevalence of impaired glucose tolerance at 1975/1976, 1982, and 1987 surveys in Nauru (sexes combined).

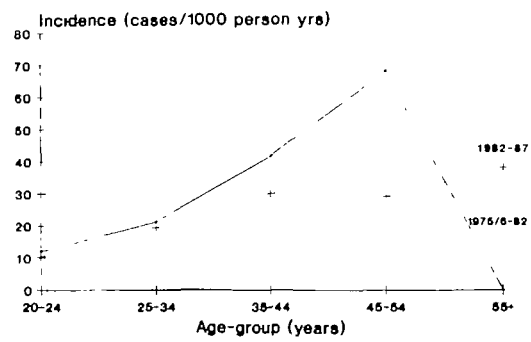

FIGURE 4. The age-specific incidence of non-insulin-dependent diabetes mellitus for the periods 1975/1976–1982 and 1982–1987 in Nauru (sexes combined).

TABLE 3. Changes in the incidence and natural history of glucose intolerance in Nauruans between 1975/1976–1982 and 1982–1987 (sexes combined)

Category and period	Person-years	Incident cases (n)	Crude incidence rate (cases/1,000 person-years)	Age-standardized incidence rate (cases/1,000 person-years)	Incidence rate ratio†	95% CI‡	χ^2_1
Normal							
IGT‡							
1975/1976–1982	876	32	36.5	41.4	0.55	0.36–0.84	7.5**
1982–1987	3,299	64	19 4	21.6			
NIDDM‡							
1975/1976–1982	876	15	17.2	17.1	0.46	0.24–0.85	6.0*
1982–1987	3,299	27	8.2	7.4			
IGT							
NIDDM							
1975/1976–1982	390	22	58.3	35.2	1.23	0.72–2.10	0.6
1982–1987	959	58	60.5	56.1			
Normal							
1975/1976–1982	390	23	59 0	69.8	1.28	0.80–2.04	1.0
1982–1987	959	77	80.3	88.1			
All							
NIDDM							
1975/1976–1982	1,266	37	29.2	26.2	0.77	0.52–1.16	1.5
1982–1987	4,258	85	20.0	22.5			

* $p < 0$ 05; ** $p < 0.01$
† Mantel-Haenszel incidence density rate ratio and test-based confidence interval (19)
‡ CI, confidence interval; IGT, impaired glucose tolerance, NIDDM, non-insulin-dependent diabetes mellitus.

TABLE 4. Risk factor levels in nondiabetic Nauruans aged ≥20 years, as determined in 1975/1976, 1982, and 1987 cross-sectional surveys

	Risk factor/glucose tolerance									
	Low physical activity (%)		Current cigarette smoking (%)		Mean body mass index (kg/m²)		Mean fasting triglycerides (mmol/liter)		Mean total cholesterol (mmol/liter)	
	Normal	IGT†	Normal	IGT	Normal	IGT	Normal	IGT	Normal	IGT
1975/1976	74.7	81.1	69.7	52.3	32.8	33.5	NS†	NS	NS	NS
1982	NA†	NA	63.4	64.0	33.3	35.8**	1.02	1.33	4.2	4.6
1987	79.6	76.8	61.5*	68.0*	34.2*	37.5***	1.11	1.30	5.0	5.4

* $p < 0$ 05; ** $p < 0.01$, *** $p < 0$ 001 Significantly different from 1975/1976 (where data available) by χ^2 or t test.
† IGT, impaired glucose tolerance; NS, not shown because methodology not comparable to 1982 and 1987 surveys; NA, not assessed.

TABLE 5. Livebirths by age and glucose tolerance in Nauruan women as reported during 1987 survey

	Age group (years)			
	20–34	35–44	45–54	55+
No. of subjects	329	141	89	99
Glucose tolerance				
Normal	3.02 ± 0.15*	4.54 ± 0.39	7.91 ± 0.64	8.22 ± 0.88
IGT†	2.34 ± 0.54	6.42 ± 0.63	4.50 ± 0.94	7.57 ± 1.20
NIDDM†	2.54 ± 0.42	4.30 ± 0.59	7.44 ± 0.50	6.92 ± 0.52
	NS†	$p < 0.01$	NS	NS

* Mean ± standard error.
† IGT, impaired glucose tolerance; NIDDM, non-insulin-dependent diabetes mellitus; NS, not significant (one-way analysis of variance across glucose tolerance categories within each age group).

ance. Similarly, women with impaired glucose tolerance have also had fewer livebirths, excepting those aged 35–44 years.

Even though the cross-sectional results of table 5 demonstrate a reduction in mean livebirths in diabetic women, it remains likely that, in many of these women (particularly those who are postmenopausal), the onset of the disease in fact occurred following some or all of their pregnancies. Assuming that an effect of glucose intolerance on number of livebirths is most likely to occur after the onset of the problem, such a cross-sectional analysis may obscure a stronger relation in younger women whose onset of non-insulin-dependent diabetes (which is also likely to be of a more severe form than that with onset later in life) is definitely prior to pregnancy.

To investigate this possibility, we have examined the number of livebirths longitudinally for the period 1982–1987 in women still in the reproductive age range (20–44 years) at baseline. After adjusting for age by analysis of covariance, those who had glucose intolerance at baseline had significantly fewer livebirths over the ensuing 5 years than did their peers with normal glucose tolerance (age-adjusted mean livebirths: normal, 1.01; impaired glucose tolerance, 0.93; non-insulin-dependent diabetes, 0.48; $p < 0.001$ by F test).

DISCUSSION

These data confirm that Micronesian Nauruans have a prevalence and incidence of diabetes mellitus (exclusively non-insulin-dependent, to date) that rank second only to those found in the Pima Indians of Arizona (5, 10). On an age-standardized basis, the incidence in Pima Indians was 38.2 cases/1,000 person-years in 1965–1975, compared with the 1982–1987 incidence of 22.5 cases/1,000 person-years in Nauruans. As demonstrated in table 6, these rates far exceed those seen in predominantly Caucasian populations in the United States (5, 20, 21), Sweden (22), Israel (23), and Malta (24), irrespective of methodological differences. The only population with a recorded incidence rate approaching those of the Pimas and Nauruans would appear to be Tokelauan (Polynesian) immigrants to New Zealand (25, 26). It is probable, however, that incidence rates intermediate between those of Pimas and Nauruans on the one hand and Caucasians on the other occur in many

TABLE 6. Estimated incidence rate for diabetes from various studies*

Authors	Population	Period	Basis for diagnosis	Age range (years)	Incidence/1,000 person-years
Present study	Nauruans (Micronesian)	1982–1987	WHO† (1985) (17)	20+	22.5‡
Knowler et al. (5)	Pima Indians (native American)	1965–1975	WHO (1985) (17)	5+	26.5 [38.2]‡
	Rochester, MN (mostly white)	1945–1970	Patient records + various blood glucose criteria	All ages	1.3
Wilson et al. (20)	Framingham, MA	1954–1968	Physician + various blood glucose criteria	33–67	4.8 males 3.9 females
Butler et al. (21)	Tecumseh, MI	1959–1979	Self-report and death certificate	20+	2.6
Ohlson et al. (22)	Swedish males	1963–1980	100-g OGTT† WHO (1982) (17)	51–60 61–67	4.3 4.9
Medalie et al. (23)	Israeli males (various ethnic groups)	1963–1968	Various blood glucose criteria	40+	8.0
Schranz (24)	Maltese	1981–1987	WHO (1985) (17)	35+	9.7
Stanhope and Prior (25)	Tokelauan (Polynesian) migrants to New Zealand	1969–1977	100-g OGTT, 1-hour plasma glucose ≥13.9 mmol/liter	20+	11.0 males 21.8 females

* Readers should beware of many caveats in comparing these studies. Consult original publications for full details.
† WHO, World Health Organization; OGTT, oral glucose tolerance test.
‡ Age standardized to Nauru 1982 survey population, for ages 20–24, 25–34, . . . 55+.

formerly traditional populations that have experienced rapid socioeconomic development and are known to have a high prevalence of non-insulin-dependent diabetes. These include Australian Aborigines (8), Asian Indians (6, 7), and other Pacific islander (1, 3) and North American Indian (4, 27) groups.

The age-standardized incidence rate of non-insulin-dependent diabetes in Nauruans for the period 1982–1987 was 14 percent lower than that found for 1975/1976–1982. However, this result was produced by two contrasting trends: a halving in the rate of development of impaired glucose tolerance or non-insulin-dependent diabetes among subjects with normal glucose tolerance, and a 60 percent increase in the conversion rate from impaired glucose tolerance to non-insulin-dependent diabetes. Combined with an increase in the rate of reversion from impaired glucose tolerance to normal and similar mortality rates for subjects with either impaired glucose tolerance or non-insulin-dependent diabetes over the two periods (unpublished data), the age-standardized prevalence of impaired glucose tolerance fell by more than half from 1982 to 1987, and the prevalence of non-insulin-dependent diabetes was constant.

The most important feature of these trends is the significant decline in the incidence of impaired glucose tolerance and non-insulin-dependent diabetes in subjects with normal glucose tolerance at baseline. The magnitude of the change and the consistency in both sexes and for both categories of glucose intolerance suggest that the change is unlikely to be due to chance. Given that 78 percent of the 1987 total population aged 25 years and above were cohort members and that 86 percent of these were seen in the follow-up survey, it would appear that any selection bias in the longitudinal study cohort would have to be large, indeed, to explain the observed trends. That the longitudinal cohort and the final study population were representative of the total Nauruan population is suggested by 1) similar age distributions (table 1); 2) cohort responders and nonresponders (including those temporarily overseas) were similar for a variety of biochemical variables measured at earlier surveys (16); 3) permanent emigration from Nauru is insignificant; 4) similar diabetes-specific mortality rates in cohort and noncohort members for the period up until the survey (unpublished data); and 5) evidence that random and fasting glucose levels in noncohort members were similar to basal levels in the cohort (16).

The latter two pieces of evidence suggest that having attended earlier surveys did not result in a favored prognosis relative to nonattenders and indicate that the decline in the incidence of impaired glucose tolerance and non-insulin-dependent diabetes in normal subjects may well have occurred for reasons other than risk factor reduction related to participation in the longitudinal studies. Furthermore, if the decline in incidence of glucose intolerance in normal subjects was due to improvements in life-style-related risk factors, then one would have expected this to have also led to a fall, rather than the observed increase, in the incidence of non-insulin-dependent diabetes from impaired glucose tolerance. (This may be a specious argument, however, in that once decompensation to impaired glucose tolerance has occurred, further progression to non-insulin-dependent diabetes could be less dependent on the level of environmental risk factors).

Indeed, risk factor levels in nondiabetic Nauruans do not appear to have improved over the course of these studies. Obesity and, to a lesser extent, physical activity are the two most widely recognized modifiable risk factors for glucose intolerance (3, 17). Our data indicate that the mean body mass index has actually increased significantly in nondiabetic Nauruans over the period of these surveys and, although our measure of physical activity is crude, it too does not indicate any improvement. We have no comparative dietary data over the period but, as indirect indicators, triglyceride and cholesterol levels appear to have increased between 1982 and 1987 in individuals with normal glucose tolerance, suggesting that any change in diet is unlikely to have been too advantageous. In light of the above, the decline in cigarette

smoking in normal subjects (it appears to have actually increased by a greater amount in those with impaired glucose tolerance) does not seem very compelling evidence for general improvements in health behavior. Although anecdotal, one of us (P.Z.Z.) has been visiting Nauru regularly since 1975 and sees little evidence for beneficial life-style change.

It it possible that differences in misclassification rates of glucose tolerance status at different surveys could have produced biased incidence estimates, but there is no reason to suspect that such differences should have occurred. Nor can the present decline be ascribed to increasing foreign genetic admixture in the population (27, 28), as the changes have occurred in all age groups and by no means solely in the younger cohort of individuals represented only in the 1982–1987 incidence calculations.

The most plausible explanation for the fall in the incidence of glucose intolerance in Nauruans may therefore be that a "common source" epidemic has already drained the most susceptible individuals from the pool of individuals with normal glucose tolerance. That is, the adverse environmental influence has been sufficiently widespread and strong that a high proportion of the genetically susceptible Nauruans had already developed impaired glucose tolerance or non-insulin-dependent diabetes prior to the 1982 survey. Hence, the frequency and/or potency of the diabetic genotype was lower in the cohort of subjects whose glucose tolerance remained normal in 1982 compared with the group who were normal in 1975/1976.

It has previously been demonstrated that transmission of non-insulin-dependent diabetes mellitus in Nauru is compatible with an autosomal dominant mode of inheritance, although the nature of the defect is unknown (29). The high frequency of the diabetes genotype in traditional populations such as Pacific islanders, Australian Aborigines, and North American Indians has been explained on the basis of Neel's hypothesis of a "thrifty genotype" (3, 13, 14).

That the genotype could have reached such high frequency in Nauru is consistent with its position as one of the smallest and more isolated of inhabited Pacific islands, implying a long and difficult canoe voyage prior to original settlement and the unlikelihood of significant genetic inflow from subsequent arrivals (28). Moreover, there is relatively little arable soil, and the island is particularly susceptible to droughts. In more recent times, World War II provided a final insult compatible with Neel's selection hypothesis. Prior to Japanese occupation in 1942, the Nauruan population numbered 1,848 persons, but approximately 500 (27%) of these died by the end of the war, the majority probably from malnutrition on the distant island of Truk, to which two thirds of the population had been forcibly deported (30). It is conceivable that the extreme conditions experienced by the Nauruans during the war may have resulted in a further selective increase in the frequency of the diabetic genotype. Certainly, routine data and medical reports indicate that it was not until the early 1950s that the present diabetes epidemic had its origin, although single cases were recorded in the mortality data for 1925 and 1935 (31).

Speculatively, the documented fall in incidence of glucose intolerance in normal subjects and in the prevalence of impaired glucose tolerance (itself the major "risk factor" for non-insulin-dependent diabetes) may herald a continuing decline in the overall incidence, and eventually in the prevalence, of non-insulin-dependent diabetes in Nauruans, related to natural selection against the genotype and dilution through continuing foreign genetic admixture. Non-insulin-dependent diabetes has an early age of onset in highly susceptible populations such as Nauruans (with a prevalence of 10–30 percent through much of the reproductive phase). These young diabetic patients are known to experience the full range of disease sequelae, including retinopathy (32), renal failure (16), and obstetric complications (33), and they have higher mortality rates than their nondiabetic peers (34). There is evidence that both impaired glucose

tolerance and non-insulin-dependent diabetes in young Nauruan women are strongly associated with stillbirth occurrence (33), and we have shown here that diabetic Nauruan women of all age groups have a lower number of livebirths than do their nondiabetic peers. Perhaps of greater importance, the longitudinal data suggest that the fertility deficit is pronounced in women with an early onset (and possibly therefore those with the more adverse genotype).

This combination of circumstances implies that the diabetic genotype (or at least its most severe form) will be transmitted with decreasing frequency in future generations, as the wheel has come full circle on the formerly advantageous "thrifty genotype." An analogous situation has been observed in diabetes-susceptible rat species, where the tendency to hyperglycemia decreases in successive generations of laboratory-raised animals (35).

The decline in the frequency of the diabetic genotype in highly susceptible populations will be slow, however, and depend on the interaction of primary prevention (life-style related) measures and the medical care system (secondary and tertiary prevention), as well as the extent of foreign genetic admixture. These factors and their impact will undoubtedly vary between populations: those with effective preventive programs and access to sophisticated medical care, and/or a desire to retain their ethnic identity, will tend to perpetuate their glucose intolerance problem; while in communities with less successful preventive programs and cultural breakdown, the problem will in time recede through natural selection and genetic dilution. Either outcome has unpalatable features.

Stern and coworkers (36, 37) have logically argued (and all public health workers would hope!) that there may be a "descending limb" to the modernization-non-insulin-dependent diabetes relation, in which high-prevalence groups correct unhealthy aspects of their new life-styles, with a resulting decline in non-insulin-dependent diabetes prevalence. They were unable, however, in their study of Mexican Americans, to determine whether the lower disease prevalence observed in the most modernized group resulted from improved life-style or variation in genetic susceptibility following foreign admixture. Certainly, in Nauruans, the documented fall in incidence does not at this stage appear to be due to either improved life-style or foreign genetic admixture.

It is unclear whether the "thrifty genotype" ever occurred in developed European and Asian populations with frequencies of a magnitude similar to those now seen in Pacific islanders and other developing populations. If it was common at some time in the past, then it may well have been selected against until the gene(s) reached a relatively low potency and constant population frequency, consistent with the currently observed pattern of disease onset in middle and older age, following completion of the reproductive phase.

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